

XYLOFY AI WEEK 1 SUMMARY

Project Summary

In this project, I built a machine learning model to predict house prices using housing data. First, I loaded and explored the dataset to understand its structure, identify the target variable (price), and check for missing values. Next, I cleaned the data by handling missing values, removing duplicate records, and converting categorical features into numerical form using one-hot encoding.

After preprocessing, I split the dataset into training and testing sets and trained two regression models: Linear Regression and Random Forest Regressor. The performance of both models was evaluated using MAE, RMSE, and R^2 Score to determine their prediction accuracy. I also created visualizations such as a house price distribution histogram, a correlation heatmap, and an actual vs. predicted price plot to better understand the data and model performance.

Finally, I analyzed the results to identify the most influential factors affecting house prices and compared the effectiveness of the two models. The project demonstrates how machine learning can be used to estimate house prices and support data-driven decision-making in the real estate industry.